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Sample Question Set - Unit 6

Course Code: CSE252
Course Title: INTRODUCTION TO ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING
Course Outcome: CO6
Paper: Sample Multiple Choice Questions - ANSWER KEY
Maximum Marks: 40 x 1 = 40

Unit: 6
Unit Title: AI for Intelligent Systems and Robotics
Unit Mapped RBT: L3 (Apply)
Total Questions: 40
Suggested Duration: 60 Minutes

Answer Grid (Key)

Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
Q1	A	Q11	D	Q21	A	Q31	C
Q2	B	Q12	B	Q22	C	Q32	A
Q3	C	Q13	D	Q23	B	Q33	A
Q4	D	Q14	C	Q24	D	Q34	D
Q5	B	Q15	B	Q25	C	Q35	B
Q6	D	Q16	A	Q26	D	Q36	C
Q7	A	Q17	B	Q27	A	Q37	C
Q8	C	Q18	A	Q28	B	Q38	B
Q9	C	Q19	D	Q29	D	Q39	D
Q10	A	Q20	C	Q30	B	Q40	A

Multiple Choice Questions (40 x 1 = 40 Marks)

Q1. Name the computer vision task that locates each object present and draws a box around it.

A) Object detection <-- correct
B) Image classification
C) Tokenisation
D) Clustering

Answer: A. Classification returns one label for the whole image; detection returns a box and a label for every object it finds.

Q2. Identify the element of a reinforcement learning problem that describes the situation the agent currently finds itself in.

A) The action
B) The state <-- correct
C) The reward
D) The policy

Answer: B. The agent observes a state, chooses an action, receives a reward and arrives in a new state, which is the loop the whole framework rests on.

Q3. State the natural language processing step that reduces words to a common dictionary base form.

A) Tokenisation
B) Vectorisation
C) Lemmatisation <-- correct
D) Detection

Answer: C. Lemmatisation maps running and ran to run using the grammar of the language, while stemming does so by cruder rules.

Q4. Name the term for the purpose that lies behind a user's message to a chatbot.

A) The entity
B) The token
C) The utterance
D) The intent <-- correct

Answer: D. The intent is what the user wants done; the entities are the particular details, such as a date or a room number, that the request carries.

Q5. Identify the reinforcement learning quantity that reduces the weight given to rewards received later.

A) The learning rate
B) The discount factor <-- correct
C) The exploration rate
D) The reward signal

Answer: B. A discount factor below one makes an immediate reward worth more than the same reward far in the future, which keeps the total finite.

Q6. State what sensor fusion means for a robot.

A) Replacing every sensor with a single camera
B) Sending the sensor readings to a remote server
C) Reducing the number of sensors the robot carries

D) Combining readings from several sensors into one estimate <-- correct

Answer: D. No single sensor is reliable in every condition, so a camera, a depth sensor and an inertial unit are combined into one estimate of the situation.

Q7. Explain why a vision model is trained on images that have been rotated, cropped and re-lit.

A) It teaches the model to tolerate the variation it will meet <-- correct

B) It reduces the number of parameters the model needs

C) It removes the need for any labelled images

D) It makes the model run faster once deployed

Answer: A. Augmentation expands a small labelled set with realistic variations, which is one of the cheapest defences against overfitting in vision.

Q8. Distinguish an intent from an entity in the design of a chatbot.

A) The entity is what is wanted, the intent is a detail

B) Both of the two describe the purpose of the message

C) The intent is what is wanted, the entity is a detail of it <-- correct

D) Both of the two describe details within the message

Answer: C. In 'book lab 3 for Friday', the intent is to make a booking while lab 3 and Friday are the entities the system must extract.

Q9. Describe what a policy is in reinforcement learning.

A) A record of every reward the agent received

B) A list of the states the environment can hold

C) A rule mapping each state to the action to take <-- correct

D) A measure of how well the agent has performed

Answer: C. Learning a good policy is the whole object of the exercise; the rewards are only the signal by which it is judged and improved.

Q10. Interpret an object detector returning a box with a confidence of 0.35 when the threshold is set at 0.5.

A) The detection is discarded as too uncertain <-- correct

B) The detection is accepted with a warning

C) The detection is accepted and the threshold changed

D) The detector must be retrained immediately

Answer: A. The threshold decides how cautious the system is. Lowering it catches more genuine objects but raises the number of false detections.

Q11. Predict what happens to a chatbot when a user phrases a familiar request in wording it has never seen.

A) The request is answered more accurately

B) The chatbot automatically learns the new phrasing

C) The chatbot converts the request into an entity

D) The intent may not be recognised at all <-- correct

Answer: D. Adding more example utterances, or matching by meaning through embeddings rather than by exact words, is what makes the system robust to paraphrase.

Q12. Recognize the vision task described: the whole photograph is given a single label such as cat or dog.

A) Object detection

B) Image classification <-- correct

C) Sensor fusion

D) Tokenisation

Answer: B. Classification answers what is in the picture; detection also answers where each thing is.

Q13. Explain why a learning robot must balance exploration against exploitation.

A) Without exploiting it never receives any reward

B) Exploration reduces the memory the robot needs

C) Exploitation removes the need for any sensors

D) Without exploring it never finds a better course of action <-- correct

Answer: D. Always taking the best action known so far locks the agent into the first adequate strategy it happens to find.

Q14. Summarize what removing stop words contributes to processing a piece of text.

A) It corrects the spelling of every word present

B) It converts each word into its dictionary form

C) It drops very common words that carry little meaning <-- correct

D) It identifies the people named in the text

Answer: C. Words such as the and of appear everywhere and rarely help distinguish one document from another, though for some tasks they do matter.

Q15. Distinguish a rule-based chatbot from one built on a learned language model.

A) One generalises from examples, the other matches patterns

B) One matches written patterns, the other generalises from examples <-- correct

C) Both of the two match written patterns exactly

D) Both of the two generalise from training examples

Answer: B. A rule-based system is predictable and easy to audit but brittle; a learned one copes with paraphrase but can produce an answer nobody intended.

Q16. Explain why the vision model of a mobile robot is often run on board rather than on a remote server.

A) The decision must be taken faster than a round trip allows <-- correct

B) The model is far too large to fit on a server

C) A server cannot process image data at all

D) On-board models need no training data

Answer: A. A robot that must stop for an obstacle cannot wait for a network round trip, and it must keep working when the connection drops.

Q17. Predict the effect on a robot of a reward that counts the dirt collected rather than the cleanliness of the floor.

A) It cleans the floor faster than it otherwise would
B) It may deposit dirt in order to collect it again <-- correct
C) It refuses to move until the floor is already clean
D) It ignores the reward and cleans correctly anyway
Answer: B. The agent maximises what is measured, not what is meant. Rewards must be written over the state of the environment, not over the agent's activity.

Q18. Describe what a fallback response is for in a conversational system.

A) It handles a message whose intent was not recognised <-- correct
B) It answers the most common question asked
C) It stores the message for later training
D) It converts the message into a set of tokens
Answer: A. A good fallback asks a clarifying question or hands over to a person rather than guessing, which is what keeps a wrong answer from being given confidently.

Q19. Explain why named entity recognition is useful inside a chatbot.

A) It decides which language the message uses
B) It measures how polite the message is
C) It counts how many words the message holds
D) It extracts the specific details the request needs <-- correct
Answer: D. Once the intent is known, the system still needs the date, the room or the name before it can act, and those are the entities.

Q20. Distinguish an episode from a step in reinforcement learning.

A) An episode is one action, a step a whole attempt
B) Both of the two describe a single action
C) A step is one action, an episode is a whole attempt <-- correct
D) Both of the two describe a complete attempt
Answer: C. An episode runs from the starting state to a terminal one, and the agent's performance is normally measured over whole episodes.

Q21. Apply the intersection over union measure to a predicted box of area 50 and a true box of area 70 whose overlap has area 30.

A) About 0.33 <-- correct
B) About 0.43
C) About 0.60
D) About 0.25
Answer: A. The union is 50 + 70 - 30 = 90, so the intersection over union is 30 / 90, approximately 0.33.

Q22. Solve the following: an agent receives rewards of 4, 2 and 8 over three successive steps with a discount factor of 0.5. Determine the discounted return.

A) 14
B) 9
C) 7 <-- correct
D) 5
Answer: C. The return is 4 + 0.5 x 2 + 0.25 x 8 = 4 + 1 + 2 = 7.

Q23. Use a bag-of-words representation over the sentences 'robot lifts box' and 'robot drops box', and determine the size of the vocabulary.

A) 6
B) 4 <-- correct
C) 3
D) 2
Answer: B. The distinct words are robot, lifts, box and drops, so the vocabulary holds four entries.

Q24. Apply a confidence threshold of 0.6 to detections scored 0.9, 0.55, 0.7 and 0.2, and determine how many are kept.

A) 1
B) 3
C) 4
D) 2 <-- correct
Answer: D. Only 0.9 and 0.7 exceed the threshold, so two detections survive.

Q25. Examine a chatbot that answers correctly for 'book a lab slot' but fails on 'I need a laboratory booking', and determine the most appropriate remedy.

A) Raise the confidence threshold of the detector
B) Reduce the number of intents the system holds
C) Add more example phrasings or match by meaning <-- correct
D) Remove the entities from the training examples
Answer: C. The intent is the same but the wording is not. Embedding-based matching, or a wider set of training utterances, closes the gap.

Q26. Compare image classification with object detection on what each one returns for a photograph.

A) Detection returns one label, classification returns boxes
B) Both of the two return a single label per image
C) Both of the two return a box for every object
D) Classification returns one label, detection returns boxes and labels <-- correct

Answer: D. A robot that must stop for an obstacle needs to know where the obstacle is, which classification alone cannot supply.

Q27. Solve the following: a robot controller has a budget of 80 milliseconds. Its perception takes 35 and its planner 25. Determine the time left for actuation.

A) 20 milliseconds <-- correct

B) 60 milliseconds

C) 10 milliseconds

D) 45 milliseconds

Answer: A. 80 - 35 - 25 = 20 milliseconds remain, which is why a heavier vision model can make the whole loop miss its deadline.

Q28. Apply the vocabulary of reinforcement learning to a vacuum robot that reads its position, drives to a spot and receives a score for dirt collected.

A) Position is the action, driving the state, score the policy

B) Position is the state, driving the action, score the reward <-- correct

C) Position is the reward, driving the state, score the action

D) Position is the policy, driving the reward, score the state

Answer: B. Mapping a task onto state, action and reward correctly is the first step in formulating it as a reinforcement learning problem.

Q29. Apply tokenisation to the sentence 'the robot moves fast' and determine the number of word tokens.

A) 3

B) 5

C) 1

D) 4 <-- correct

Answer: D. Tokenisation splits the text into its units; here the four words are the, robot, moves and fast.

Q30. Examine a robot detector trained only on daylight photographs that fails after dark, and determine the cause.

A) The detector has too few parameters to work

B) The deployment conditions were absent from the training data <-- correct

C) The confidence threshold was set too low

D) The images were stored in the wrong format

Answer: B. A model reproduces the conditions of its training set. Collecting night images, or augmenting the existing ones, is the correction.

Q31. Compute the number of frames a detector can process each second when one inference takes 25 milliseconds.

A) 25

B) 4

C) 40 <-- correct

D) 400

Answer: C. One second holds 1000 / 25 = 40 inferences, so the detector runs at 40 frames per second.

Q32. Apply transfer learning to a robot vision task for which only 300 labelled images exist, and determine the usual approach.

A) Reuse a pre-trained network and retrain its final layers <-- correct

B) Train a new network from the beginning on the 300

C) Abandon vision and use a rule-based method

D) Collect no more data and accept the poor result

Answer: A. The early layers hold general visual features that transfer; only the task-specific head needs the small labelled set.

Q33. Use an exploration rate of 0.1 and determine the proportion of actions the agent takes at random.

A) 10 per cent <-- correct

B) 90 per cent

C) 1 per cent

D) 50 per cent

Answer: A. The remaining ninety per cent of the time the agent takes the action it currently believes to be best.

Q34. Compare running a vision model on board a robot with running it on a remote server, on latency.

A) The server is faster, since no model must be loaded

B) The two have identical latency in every case

C) Latency does not depend on where the model runs

D) On board is faster, since no network round trip is needed <-- correct

Answer: D. The server may have far more computing power, but for a safety decision the round trip usually costs more than the extra speed gains.

Q35. Examine a reinforcement learning agent that keeps repeating the first adequate action it discovered, and determine the cause.

A) Its discount factor is set above one

B) Its exploration rate is too low <-- correct

C) Its state representation is too detailed

D) Its reward signal is delivered too quickly

Answer: B. Without exploration the agent never gathers evidence about alternatives, so it settles permanently on the first strategy that worked.

Q36. Apply the design of a reward to a delivery robot and determine the better of two definitions.

A) A score for every metre the robot travels

B) A score for every second the robot is switched on

C) A score for parcels delivered on time, with a penalty for collisions <-- correct

D) A score for the number of routes it considers

Answer: C. The reward must describe the outcome wanted in the world. Rewarding distance or activity produces a robot that moves a great deal and delivers little.

Q37. Solve the following: an agent receives a reward of 10 now and 10 at the next step with a discount factor of 0.9. Determine the discounted return.

A) 20

B) 18

C) 19 <-- correct

D) 10

Answer: C. The return is 10 + 0.9 x 10 = 10 + 9 = 19.

Q38. Use an intersection over union threshold of 0.5 to determine whether a detection scoring 0.42 counts as correct.

A) It counts, since any overlap is sufficient

B) It does not count, since it falls below the threshold <-- correct

C) It counts, since 0.42 is close to 0.5

D) The threshold does not apply to detections

Answer: B. The threshold fixes how closely the predicted box must match the true one before the detection is credited.

Q39. Demonstrate how a well-designed chatbot should handle a request whose intent it cannot recognise.

A) Answer with the most common intent it holds

B) Repeat the user's message back unchanged

C) Close the conversation without any reply

D) Say so, ask a clarifying question or hand over to a person <-- correct

Answer: D. Guessing produces a confident wrong answer, which damages trust far more than an honest admission that the request was not understood.

Q40. Relate the resolution of the camera images to the computation an object detector requires.

A) Higher resolution costs more computation per frame <-- correct

B) Higher resolution costs less computation per frame

C) Resolution has no bearing on the computation

D) Higher resolution removes the need for a threshold

Answer: A. More pixels means more work per inference, so the resolution has to be traded against the frame rate the control loop demands.

----- End of Sample Question Set -----